Active Inference and the Free Energy Principle: Mathematical Foundations for Agentic AI

Research Notes — ETH Zurich ML Lecture Series

Computational Neuroscience & Machine Learning

Spring 2026

Abstract

These notes provide a mathematically rigorous introduction to the Free Energy Principle (FEP) and Active Inference, a unified framework originating in computational neuroscience that casts perception, learning, and action as variational inference. We derive variational free energy from first principles, formalize Markov blankets and their controversies, develop the Expected Free Energy objective for planning, and connect these ideas to modern agentic AI—including reinforcement learning, large language models, and multi-agent systems. We present the framework's critics fairly, survey open problems, and provide an annotated bibliography for further study.

Contents

1 Introduction and Historical Context

The **Free Energy Principle** (FEP), developed primarily by Karl Friston at University College London beginning in the mid-2000s, proposes that any self-organizing system that persists in a changing environment must minimize a quantity called *variational free energy*—or equivalently, maximize Bayesian model evidence. From this single imperative, one can derive perception (updating beliefs about the world), action (changing the world to match predictions), learning (adjusting generative model parameters), and attention (adjusting the precision of predictions).

Active Inference (AIF) is the process theory that operationalizes the FEP. Where the FEP is a principle about what self-organizing systems *must* do, active inference specifies *how* they do it: by selecting policies (action sequences) that minimize *expected free energy*, a functional that naturally balances exploration and exploitation.

Why should ML researchers care?

1. Active inference provides a principled alternative to reward-based RL, deriving curiosity-driven exploration from first principles rather than bolting it on as an auxiliary loss.
2. The mathematics connects directly to variational autoencoders (VAEs), Bayesian neural networks, and planning-as-inference.
3. The framework offers a natural language for multi-agent coordination, hierarchical planning, and embodied AI.
4. Recent work on Bayesian mechanics (Ramstead, Da Costa et al.) is building a rigorous mathematical physics of belief-updating systems.

Outline. Section **??** derives variational free energy. Section **??** covers Markov blankets and their controversies. Section **??** develops active inference and expected free energy. Section **??** introduces sophisticated inference. Section **??** covers Bayesian mechanics. Section **??** surveys generative models. Section **??** draws connections to modern agentic AI. Section **??** presents critiques. Section **??** lists open problems. Section **??** provides an annotated bibliography.

2 Variational Free Energy

2.1 Derivation from Bayes' Theorem

We begin with a generative model $p(o, \theta)$ that specifies a joint distribution over observations o and latent variables (hidden states, parameters, etc.) θ. Exact Bayesian inference requires computing:

$$p(\theta \mid o) = \frac{p(o \mid \theta)\, p(\theta)}{p(o)}, \qquad p(o) = \int p(o \mid \theta)\, p(\theta)\, d\theta. \tag{1}$$

The marginal likelihood (model evidence) $p(o)$ is typically intractable.

Step 1: Introduce a recognition density. Let $q(\theta)$ be an approximate posterior (recognition density) over θ. Consider the log model evidence:

$$\ln p(o) = \ln \int p(o, \theta)\, d\theta. \tag{2}$$

Step 2: Apply Jensen's inequality (or direct decomposition). We can decompose $\ln p(o)$ exactly. For any $q(\theta)$:

$$\begin{aligned}
\ln p(o) &= \int q(\theta) \ln p(o)\, d\theta \\
&= \int q(\theta) \ln \frac{p(o,\theta)}{p(\theta \mid o)}\, d\theta \\
&= \int q(\theta) \ln \frac{p(o,\theta)}{q(\theta)} \cdot \frac{q(\theta)}{p(\theta \mid o)}\, d\theta \\
&= \underbrace{\int q(\theta) \ln \frac{p(o,\theta)}{q(\theta)}\, d\theta}_{-\mathcal{F}[q,o] \text{ (negative free energy = ELBO)}} + \underbrace{\int q(\theta) \ln \frac{q(\theta)}{p(\theta \mid o)}\, d\theta}_{\mathrm{D_{KL}}[q(\theta)\|p(\theta|o)] \geq 0}.
\end{aligned} \tag{3}$$

Definition 2.1 (Variational Free Energy)**.** The **variational free energy** is defined as:

$$\boxed{\mathcal{F}[q,o] \;=\; -\int q(\theta) \ln \frac{p(o,\theta)}{q(\theta)}\, d\theta \;=\; \mathrm{D_{KL}}\big[q(\theta) \,\|\, p(\theta)\big] - \mathbb{E}_q\big[\ln p(o \mid \theta)\big].} \tag{4}$$

Step 3: The two decompositions. Equation (**??**) admits two useful decompositions:

(a) Complexity − Accuracy:

$$\mathcal{F} = \underbrace{\mathrm{D_{KL}}[q(\theta)\|p(\theta)]}_{\text{Complexity}} - \underbrace{\mathbb{E}_q[\ln p(o \mid \theta)]}_{\text{Accuracy}}. \tag{5}$$

Minimizing $\mathcal{F}$ balances fitting the data (accuracy) against deviating from the prior (complexity). This is Occam's razor in variational form.

(b) Energy − Entropy:

$$\mathcal{F} = \underbrace{\mathbb{E}_q[-\ln p(o,\theta)]}_{\text{Energy}} - \underbrace{H[q(\theta)]}_{\text{Entropy}}, \tag{6}$$

where $H[q] = -\mathbb{E}_q[\ln q(\theta)]$. This form connects to statistical physics: $\mathcal{F}$ is literally a free energy in the thermodynamic sense when q is interpreted as a Boltzmann distribution.

Step 4: Connection to model evidence. From (**??**):

$$\mathcal{F} = -\ln p(o) + \mathrm{D_{KL}}[q(\theta)\|p(\theta \mid o)]. \tag{7}$$

Since $\mathrm{D_{KL}} \geq 0$, we have $\mathcal{F} \geq -\ln p(o)$, i.e., $-\mathcal{F} \leq \ln p(o)$ (the ELBO). Minimizing $\mathcal{F}$ over q simultaneously: (i) tightens the bound, and (ii) drives $q(\theta) \to p(\theta \mid o)$.

2.2 Connection to Variational Autoencoders

The VAE objective is precisely the negative free energy (the ELBO):

$$\mathcal{L}_{\mathrm{VAE}}(\phi,\psi;o) = \mathbb{E}_{q_\phi(z|o)}[\ln p_\psi(o \mid z)] - \mathrm{D_{KL}}[q_\phi(z \mid o)\|p(z)] = -\mathcal{F}. \tag{8}$$

The encoder q_ϕ is the recognition model; the decoder p_ψ is the generative model. Active inference extends this by closing the loop: the "decoder" generates predictions about sensory data, and the agent *acts* to make those predictions come true.

2.3 Free Energy Minimization as a Universal Objective

Proposition 2.2. *Under the FEP, the following are all instances of free energy minimization:*

1. ***Perception:*** $q^* = \arg\min_q \mathcal{F}[q, o]$ *(update beliefs given observations).*
2. ***Action:*** $a^* = \arg\min_a \mathcal{F}[q, o(a)]$ *(change observations to reduce surprise).*
3. ***Learning:*** $\psi^* = \arg\min_\psi \mathcal{F}[q, o; \psi]$ *(adjust generative model parameters).*
4. ***Attention:*** *Optimize the precision (inverse variance) of prediction errors.*

3 Markov Blankets

3.1 Definition and Partition

Definition 3.1 (Markov Blanket)**.** Given a set of random variables $x = (\mu, s, a, \eta)$, the **Markov blanket** of the internal states μ is the set $b = (s, a)$ such that μ and η are conditionally independent given b:

$$\mu \perp\!\!\!\perp \eta \mid b = (s, a). \tag{9}$$

Here:

- μ = internal states (the system's "beliefs"),
- s = sensory states (blanket states influenced by external states),
- a = active states (blanket states influenced by internal states),
- η = external (environmental) states.

The Markov blanket induces a statistical boundary: internal states can only "know about" external states through sensory states, and can only "influence" external states through active states.

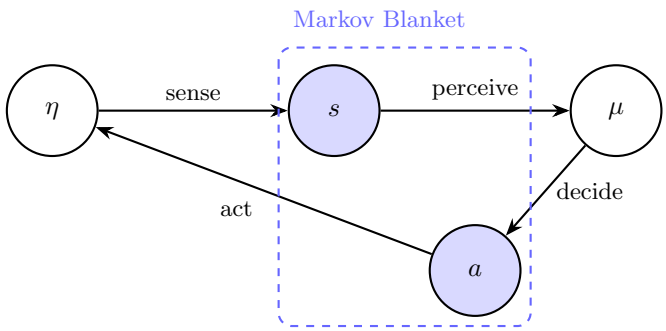

Figure 1: The four-way partition $x = (\mu, s, a, \eta)$ with the Markov blanket $b = (s, a)$ separating internal from external states.

3.2 Langevin Dynamics and the FEP

Friston's formulation starts with a Langevin (stochastic differential) equation on the full state space:

$$\dot{x} = f(x) + \omega, \qquad \omega \sim \mathcal{N}(0, 2\Gamma), \tag{10}$$

where $f(x)$ is the flow and Γ is the diffusion tensor. At the nonequilibrium steady state (NESS), the dynamics admit a decomposition:

$$f(x) = (\Gamma - Q)\, \nabla \ln p(x), \tag{11}$$

where $Q = -Q^\top$ is a solenoidal (antisymmetric) matrix encoding circulation, and Γ is the symmetric part encoding dissipation. This is the **Helmholtz decomposition** of the flow.

Given the Markov blanket partition, one can show that internal states μ parameterize a density $q_\mu(\eta) \approx p(\eta \mid b)$, so that the internal dynamics effectively perform (approximate) Bayesian inference on external states. This is the core claim of the FEP: the internal states of any Markov-blanketed system *look as if* they are performing inference.

3.3 The Controversy: Critiques of Markov Blankets under the FEP

The application of Markov blankets in the FEP has generated significant debate. We present both sides.

The Critiques.

1. **Biehl et al. (2021):** "A Technical Critique of Some Parts of the Free Energy Principle." The authors argue that:
   - The claim that internal states parameterize a density over external states relies on an unjustified identification between the *mode* of the conditional distribution $p(\eta \mid b)$ and a "belief."
   - The Markov blanket formalism in continuous-time dynamics does not straightforwardly yield the conditional independence structure assumed in graphical model theory.
   - The "as if" inference claim is not falsifiable: any system with a stationary distribution can be redescribed in these terms.
2. **Bruineberg et al. (2022):** "The Emperor's New Markov Blankets" argues that:
   - There is a conflation between *Pearl blankets* (conditional independence in a Bayesian network) and *Friston blankets* (partitions of a Langevin system at NESS).
   - Pearl blankets are properties of a graph; Friston blankets are properties of a dynamical system. The two do not trivially map onto each other.
   - The claim that "everything has a Markov blanket" is either trivially true (any partition defines conditional independence structure) or substantively false (not every system has the right kind of partition).

The Defense.

1. **Friston, Da Costa, and Parr** have responded that the critics conflate different levels of description. The FEP does not claim that every system literally performs Bayesian inference, but rather that its dynamics can be *described as if* it does—a modeling stance, not an ontological claim.
2. **Da Costa et al. (2021)** formalized Bayesian mechanics on nonequilibrium steady-state systems, providing conditions under which the "as if" interpretation holds rigorously. They distinguish between a *particular partition* (the Markov blanket exists as a dynamical structure) and the interpretation layer.
3. **Ramstead et al. (2023)** introduced the concept of a *sparse coupling* condition that replaces the graphical-model Markov blanket with a dynamical-systems notion, partially addressing Bruineberg et al.'s concerns.

Remark 3.2. For the working ML researcher, the practical takeaway is: the Markov blanket formalism is a powerful *modeling assumption* for designing agents. Whether it constitutes a "law of nature" is a philosophical question that does not affect the utility of active inference as an algorithm.

4 Active Inference

4.1 Perception and Action as Inference

In active inference, an agent has a generative model $p(o_{1:T}, s_{1:T}, \pi)$ over observations o, hidden states s, and policies π (sequences of actions). At each time step, the agent:

1. **Perceives:** Updates beliefs over hidden states by minimizing variational free energy given new observations:
$$q(s_t) = \arg\min_q \mathcal{F}[q, o_t]. \tag{12}$$

2. **Plans:** Evaluates policies by computing expected free energy (see below) and selects actions accordingly.

3. **Acts:** Executes the action prescribed by the selected policy, generating new observations.

4. **Learns:** Updates model parameters to reduce long-run free energy.

4.2 Expected Free Energy

The key innovation of active inference for planning is the **Expected Free Energy** (EFE), denoted $\mathcal{G}(\pi)$.

Definition 4.1 (Expected Free Energy)**.** For a policy π, the expected free energy at future time τ is:
$$\mathcal{G}(\pi, \tau) = \mathbb{E}_{\tilde{q}}\big[\ln q(s_\tau \mid \pi) - \ln p(o_\tau, s_\tau \mid \pi)\big], \tag{13}$$
where $\tilde{q} = q(o_\tau, s_\tau \mid \pi)$ is the predictive distribution under the approximate posterior.

Decomposition into epistemic and pragmatic value.

$$\mathcal{G}(\pi, \tau) = \underbrace{-\mathbb{E}_{\tilde{q}}\big[\mathrm{D}_{\mathrm{KL}}[q(s_\tau \mid o_\tau, \pi) \| q(s_\tau \mid \pi)]\big]}_{\text{Epistemic value (neg. information gain)}} + \underbrace{\mathbb{E}_{\tilde{q}}\big[\ln q(o_\tau \mid \pi) - \ln p(o_\tau)\big]}_{\text{Pragmatic value (neg. expected log preference)}}.$$

Rewriting with clearer signs:

$$\boxed{\mathcal{G}(\pi) = -\underbrace{\text{Information Gain}}_{\text{Exploration}} - \underbrace{\text{Pragmatic Value}}_{\text{Exploitation}}.} \tag{14}$$

- **Information Gain** (epistemic value): the expected reduction in uncertainty about hidden states after observing the outcome of a policy. This drives *exploration*—the agent seeks out informative observations.

- **Pragmatic Value**: how well the predicted observations align with the agent's preferred observations $p(o_\tau)$ (prior preferences). This drives *exploitation*—the agent pursues goals.

Policy selection. Policies are selected via a softmax over negative EFE:

$$p(\pi) = \sigma(-\gamma \mathcal{G}(\pi)) = \frac{\exp(-\gamma \mathcal{G}(\pi))}{\sum_{\pi'} \exp(-\gamma \mathcal{G}(\pi'))}, \tag{15}$$

where $\gamma > 0$ is a precision (inverse temperature) parameter.

Aspect	Active Inference	Reinforcement Learning
Objective	Minimize expected free energy	Maximize expected cumulative reward
Exploration	Intrinsic (information gain)	Extrinsic (epsilon-greedy, UCB, etc.)
Reward	Prior preferences over observations	Externally defined reward function
Model	Generative model (required)	Optional (model-free or model-based)
Optimality	Bayes-optimal given model	Bellman-optimal given MDP
Curiosity	First-class citizen	Auxiliary bonus (e.g., ICM, RND)
Multi-objective	Natural (prior preferences)	Requires reward shaping

Table 1: Active inference vs. reinforcement learning.

4.3 Contrast with Reinforcement Learning

Remark 4.2. Active inference and RL are not incompatible. When information gain is zero (no epistemic uncertainty) and we identify $\ln p(o) = r(o)$, the EFE reduces to expected reward, and active inference recovers KL-regularized RL (similar to SAC/MaxEnt RL). The key difference is that exploration is derived, not added.

4.4 The Full Active Inference Loop

Algorithm 1 Discrete Active Inference Agent (one time step)

Require: Generative model $p(o_{1:T}, s_{1:T}, \pi)$, current beliefs $q(s_t)$

Observe o_t
Perception: Update $q(s_t) \leftarrow \arg\min_q \mathcal{F}[q, o_t]$ ▷ State estimation
for each policy $\pi \in \Pi$ **do**
  Compute $\mathcal{G}(\pi) = \sum_{\tau > t} \mathcal{G}(\pi, \tau)$ ▷ Expected free energy
end for
Policy selection: Sample $\pi^* \sim \sigma(-\gamma\, \mathcal{G}(\pi))$
Action: Execute $a_t = \pi^*(t)$ ▷ First action of selected policy
Learning: Update model parameters to reduce $\mathcal{F}$ ▷ Optional

5 Sophisticated Inference

Standard active inference evaluates policies by summing EFE over future time steps, treating each step independently. **Sophisticated inference** (Friston et al., 2021) extends this by introducing *recursive policy evaluation*: the agent considers what it would *believe and do* in the future, given possible observations.

5.1 Recursive Evaluation

At each decision point, the sophisticated agent:

1. Generates possible future observations $o_{\tau+1}$ under each policy.
2. For each possible observation, computes the updated beliefs $q(s_{\tau+1} \mid o_{\tau+1})$.
3. Given those updated beliefs, recursively evaluates what policy would be selected at time $\tau + 1$, and so on.
4. The expected free energy is computed over the full tree of future contingencies.

Formally, the sophisticated EFE at depth d is:

$$\mathcal{G}^{(d)}(\pi, \tau) = \mathcal{G}(\pi, \tau) + \mathbb{E}_{\tilde{q}(o_{\tau+1}|\pi)}\big[\min_{\pi'} \mathcal{G}^{(d-1)}(\pi', \tau + 1 \mid o_{\tau+1})\big], \tag{16}$$

with $\mathcal{G}^{(0)} = \mathcal{G}$ (the standard EFE).

5.2 Connection to MCTS and AlphaZero

Sophisticated inference bears a structural resemblance to Monte Carlo Tree Search (MCTS):

- Both build a tree of future contingencies.
- Both evaluate leaf nodes with a value function (EFE plays this role).
- Both propagate values back up the tree.

The key differences:

- MCTS uses rollouts or a learned value network; sophisticated inference uses the generative model's predictive distributions.
- AlphaZero's value/policy networks can be seen as amortized versions of the recursive EFE computation.
- Sophisticated inference naturally handles partial observability (POMDPs), while standard MCTS typically assumes full state observability.

Remark 5.1. Champion et al. (2022) have shown that deep active inference agents using tree search can achieve competitive performance with AlphaZero-style agents in certain domains, while requiring less training data due to the built-in epistemic drive.

6 Bayesian Mechanics

Bayesian mechanics is a recent program (Ramstead, Da Costa, Sakthivadivel, Friston et al., 2023–2025) that aims to place the FEP on rigorous mathematical-physics foundations.

6.1 The Core Idea

Bayesian mechanics studies the mechanics of systems that can be described as performing inference—specifically, systems whose internal states parameterize probability distributions over external states. The central question: given a particular partition (a Markov blanket structure) of a dynamical system, what are the equations of motion for the "beliefs" encoded by internal states?

6.2 Path Integral Formulation

Following the Feynman–Kac approach, one can express the marginal likelihood over a trajectory of observations $o_{0:T}$ as a path integral:

$$p(o_{0:T}) = \int \mathcal{D}[s_{0:T}]\, p(o_{0:T} \mid s_{0:T})\, p(s_{0:T}), \tag{17}$$

where $\mathcal{D}[s_{0:T}]$ denotes integration over all possible hidden state trajectories. The variational free energy then becomes a functional over trajectory space:

$$\mathcal{F}[q] = \mathrm{D}_{\mathrm{KL}}\big[q(s_{0:T}) \,\|\, p(s_{0:T})\big] - \mathbb{E}_q\big[\ln p(o_{0:T} \mid s_{0:T})\big]. \tag{18}$$

6.3 Connection to Jaynes' Maximum Entropy

Jaynes' MaxEnt principle states that the least biased distribution consistent with known constraints is the one with maximum entropy. Bayesian mechanics connects this to the FEP:

Proposition 6.1. *At the NESS of a Langevin system, the steady-state density $p_\infty(x)$ maximizes entropy subject to the constraint that the expected energy is fixed. The internal states' dynamics then minimize a free energy functional that is formally identical to Jaynes' MaxEnt objective.*

This provides a deep link between information geometry, statistical physics, and the FEP.

6.4 Hamilton's Equations for Belief Systems

Da Costa and colleagues have shown that, under certain regularity conditions, the dynamics of belief-parameterizing internal states can be written in Hamiltonian form:

$$\dot{\mu}_i = \frac{\partial \mathcal{H}}{\partial \nu_i}, \qquad \dot{\nu}_i = -\frac{\partial \mathcal{H}}{\partial \mu_i} + \text{dissipation}, \tag{19}$$

where μ_i are the "belief coordinates," ν_i are conjugate momenta, and $\mathcal{H}$ is a Hamiltonian that plays the role of free energy. The dissipation term reflects the irreversible, entropy-producing character of inference in open systems.

Remark 6.2. This is a genuinely new result: it says that belief dynamics obey a generalization of Hamilton's equations from classical mechanics, providing a "physics of inference." The solenoidal (conservative) part drives oscillations and cycles in belief space; the dissipative part drives convergence to the posterior.

7 Generative Models for Active Inference

Active inference is model-based by construction. The choice of generative model determines the agent's representational capacity and computational requirements.

7.1 Discrete State-Space Models

Hidden Markov Models (HMMs). The simplest generative model:

$$p(s_t \mid s_{t-1}) = \mathbf{B}\,\delta_{s_{t-1}}, \qquad \text{(transition model)} \tag{20}$$

$$p(o_t \mid s_t) = \mathbf{A}\,\delta_{s_t}, \qquad \text{(observation model)} \tag{21}$$

where $\mathbf{A}$ and $\mathbf{B}$ are categorical probability matrices. Perception reduces to Bayesian filtering; planning to evaluating a finite set of policies.

POMDPs. Active inference POMDPs extend HMMs with policy-dependent transitions:

$$p(o_{1:T}, s_{1:T} \mid \pi) = p(s_1) \prod_{t=1}^{T} p(o_t \mid s_t) \prod_{t=2}^{T} p(s_t \mid s_{t-1}, \pi(t-1)). \tag{22}$$

The agent maintains beliefs $q(s_t \mid \pi)$ and selects policies by minimizing EFE. This is the formulation implemented in the `pymdp` library (Heins et al., 2022).

7.2 Continuous State-Space Models

For continuous domains, the generative model takes the form of stochastic differential equations:

$$ds = f(s, a)\, dt + \sigma_s\, dW_s, \tag{23}$$

$$o = g(s) + \sigma_o\, \epsilon, \tag{24}$$

where f is the state dynamics, g is the observation function, and W_s, ϵ are Wiener processes / noise. Inference proceeds via **generalized filtering** or **predictive coding**, which can be seen as gradient descent on free energy in generalized coordinates of motion (position, velocity, acceleration, ...).

7.3 Deep Active Inference

Recent work replaces the matrices $\mathbf{A}$, $\mathbf{B}$ with neural networks:

- **Deep generative models:** The observation model $p_\psi(o \mid s)$ is a neural network decoder; the recognition model $q_\phi(s \mid o)$ is a neural network encoder. This yields a VAE-like architecture embedded in an active inference loop.
- **Transition models:** $p_\xi(s_{t+1} \mid s_t, a_t)$ can be a recurrent or transformer-based model ("world model").
- **Policy networks:** $q_\omega(\pi)$ amortizes policy selection, analogous to the actor in actor-critic methods.

7.4 Hybrid and Hierarchical Models

Pezzulo and colleagues have developed hierarchical active inference models where:

- Higher levels operate on slower timescales and more abstract state spaces.
- Lower levels operate on faster timescales and concrete sensorimotor representations.
- Each level minimizes its own free energy, with top-down predictions and bottom-up prediction errors propagating between levels.

This mirrors the hierarchical structure of the cortex and provides a principled approach to temporal abstraction (cf. options in hierarchical RL).

8 Connections to Modern Agentic AI

8.1 Active Inference vs. Deep RL

Beyond the conceptual differences in Table **??**, recent empirical work has compared active inference agents with RL baselines:

- In structured, partially observable environments, active inference agents tend to learn faster due to epistemic drive (built-in curiosity).
- In high-dimensional environments with large action spaces, deep RL agents currently outperform active inference agents due to better scaling of function approximation.
- Hybrid approaches (e.g., using EFE as an intrinsic reward in an RL framework) show promise.

8.2 Large Language Models Through the Free Energy Lens

LLMs can be interpreted through the FEP framework:

- **Next-token prediction is free energy minimization:** The training objective $-\sum_t \ln p_\theta(x_t \mid x_{<t})$ is the negative log-likelihood, which is the accuracy term of free energy (with implicit complexity regularization via weight decay, dropout, etc.).
- **In-context learning as inference:** The context window acts as a form of approximate Bayesian inference—the model implicitly conditions on context to update its "beliefs" about the task.
- **Agentic LLMs:** When LLMs are equipped with tools, memory, and planning capabilities, they begin to resemble active inference agents with deep generative models. The key missing ingredient is typically the epistemic drive—current agentic LLMs do not systematically seek information to reduce uncertainty.

8.3 Robotics and Embodied AI

Active inference has been applied to:

- Robot navigation and SLAM (simultaneous localization and mapping), where the epistemic drive naturally leads to frontier exploration.
- Manipulation tasks, where predictive coding provides a natural framework for sensorimotor integration.
- Soft robotics, where the continuous-time formulation of active inference maps naturally onto flexible body dynamics.

8.4 Multi-Agent Active Inference

When multiple active inference agents interact, each agent's Markov blanket is nested within a larger system. This leads to:

- **Shared generative models:** Agents that model each other's beliefs (theory of mind as inference).
- **Collective free energy minimization:** Groups of agents can be described as a single "super-agent" minimizing collective free energy.
- **Communication as active inference:** Language and signaling emerge as actions that reduce joint uncertainty.

8.5 pymdp: A Python Library for Active Inference

`pymdp` (Heins, Millidge, Da Costa et al., 2022) is an open-source Python package for simulating active inference agents with discrete state-space models. Key features:

- Implements the full POMDP-based active inference loop.
- Supports multi-factor state spaces and hierarchical models.
- Computes EFE and policy selection.
- Provides tools for learning model parameters.
- Built on NumPy/JAX for efficient tensor operations.

A minimal pymdp agent:

```
import pymdp
from pymdp.agent import Agent
from pymdp import utils

# Define generative model
A = utils.random_A_matrix(num_obs=[5], num_states=[3])
B = utils.random_B_matrix(num_states=[3], num_controls=[3])
C = utils.obj_array_zeros(num_obs=[5])  # prior preferences
C[0][0] = 1.0  # prefer observation 0

# Create agent
agent = Agent(A=A, B=B, C=C, policy_len=2)

# Perception-action loop
obs = env.reset()
for t in range(T):
    q_s = agent.infer_states(obs)
    q_pi, G = agent.infer_policies()
    action = agent.sample_action()
    obs = env.step(action)
    agent.update_params()
```

9 Controversies and Critiques

The FEP and active inference have attracted both enthusiastic adoption and pointed criticism. A balanced assessment requires engaging with both.

9.1 Unfalsifiability

The critique. The FEP has been described as "unfalsifiable" because it applies to *any* system that maintains itself over time. If everything minimizes free energy, then the principle cannot be tested—it is tautological.

The response. Friston and colleagues distinguish between the *principle* (which is indeed a tautology, like the principle of least action in physics) and the *process theory* (active inference), which makes specific, testable predictions about neural dynamics, behavior, and psychophysics. The FEP constrains the space of possible process theories; active inference is one such theory.

Assessment. This is analogous to the status of the second law of thermodynamics: saying "entropy increases" is not falsifiable, but specific predictions about heat engines *are*. The FEP is best understood as a framework, not a hypothesis.

9.2 "It's Just Variational Inference"

The critique. Some ML researchers argue that active inference is "just" variational inference with extra steps, and that the neuroscience framing adds unnecessary complexity without improving performance over standard model-based RL.

The response.

- The EFE provides a principled derivation of intrinsic motivation, which must be manually designed in RL.
- The framework unifies perception, action, and learning under a single objective, rather than requiring separate loss functions.
- The "just variational inference" critique applies equally to VAEs, which are nonetheless highly useful.

9.3 The Markov Blanket Debate

See Section **??** for a detailed treatment. The core issue is whether continuous-time dynamical systems genuinely instantiate the conditional independence structure required for the "inference" interpretation, or whether the Markov blanket formalism is being stretched beyond its mathematical foundations.

9.4 Consciousness and Life

Some proponents have used the FEP to make strong claims about consciousness and the nature of life (e.g., that all living systems are Markov-blanketed inference machines). Critics argue that this conflates a useful modeling framework with a metaphysical theory, risking panpsychist or vitalist overreach.

9.5 Scalability

Active inference agents, particularly those using explicit EFE computation, face combinatorial scaling in the number of policies. Deep active inference addresses this through amortization, but currently lags behind deep RL in terms of scalability to complex environments.

10 Open Problems

1. **Scalable deep active inference.** Current deep active inference methods do not match the performance of state-of-the-art deep RL in high-dimensional environments (e.g., Atari, continuous control benchmarks). How can we scale EFE-based planning to large action and observation spaces without losing the principled exploration-exploitation balance?
2. **Structure learning.** Active inference agents typically assume a fixed generative model structure. How should agents learn the *structure* (number of hidden states, factorization, hierarchical depth) of their generative models online? This connects to Bayesian model comparison and neural architecture search.
3. **Continuous-time active inference with deep models.** The continuous-time formulation (generalized filtering, predictive coding) has not been successfully combined with deep neural network function approximators. Neural ODEs and diffusion models offer promising directions.
4. **Multi-agent active inference at scale.** While the theory of multi-agent active inference is elegant, practical implementations with many agents face challenges in belief propagation, communication, and scalability. Can active inference provide advantages over multi-agent RL in cooperative and competitive settings?

5. **Formal guarantees.** Under what conditions does active inference converge? What are the regret bounds? How does the quality of the generative model affect performance? The theoretical foundations need strengthening to match the maturity of RL theory.

6. **Integration with foundation models.** How can active inference principles be integrated into LLM-based agents? Can EFE or information gain be computed in the latent spaces of large pre-trained models to guide agentic behavior?

7. **Rigorous Bayesian mechanics.** The program of Bayesian mechanics is still in its early stages. Key open questions include: when exactly does the Hamiltonian formulation hold? What is the role of gauge freedom? Can the formalism handle systems far from steady state?

8. **Bridging Pearl and Friston blankets.** The Markov blanket debate has highlighted genuine mathematical gaps. A satisfactory resolution would require a formal theory connecting graphical model conditional independence with dynamical systems partitions, possibly through the lens of algebraic topology or category theory.

11 Annotated Bibliography

[1] **Friston, K. (2010).** "The Free-Energy Principle: A Unified Brain Theory?" *Nature Reviews Neuroscience*, 11(2), 127–138.
The foundational paper presenting the FEP as a unifying principle for brain function. Accessible overview; start here.

[2] **Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., & Pezzulo, G. (2017).** "Active Inference: A Process Theory." *Neural Computation*, 29(1), 1–49.
The definitive process-theory paper. Derives the expected free energy functional and connects active inference to RL and optimal control.

[3] **Parr, T., Pezzulo, G., & Friston, K. (2022).** *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior.* MIT Press.
The textbook. Comprehensive, mathematically rigorous, and well-organized. Essential reading for anyone serious about the field.

[4] **Da Costa, L., Parr, T., Sajid, N., Veselic, S., Neacsu, V., & Friston, K. (2020).** "Active Inference on Discrete State-Spaces: A Synthesis." *Journal of Mathematical Psychology*, 99, 102447.
Clear mathematical treatment of discrete active inference (POMDPs). Good companion to the pymdp documentation.

[5] **Heins, C., Millidge, B., Da Costa, L., et al. (2022).** "pymdp: A Python Library for Active Inference in Discrete State Spaces." *Journal of Open Source Software*, 7(73), 4098.
The pymdp library paper. Essential for anyone wanting to implement active inference agents.

[6] **Friston, K., Da Costa, L., Hafner, D., Hesp, C., & Parr, T. (2021).** "Sophisticated Inference." *Neural Computation*, 33(3), 713–763.
Introduces recursive policy evaluation. Connects active inference to tree search and MCTS.

[7] **Biehl, M., Pollock, F. A., & Kanai, R. (2021).** "A Technical Critique of Some Parts of the Free Energy Principle." *Entropy*, 23(3), 293.
Important critical paper. Challenges the mathematical underpinnings of the FEP, particularly the "as if" inference claim.

[8] **Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022).** "The Emperor's New Markov Blankets." *Behavioral and Brain Sciences*, 45, e183.
Influential critique of Markov blanket usage in the FEP. Distinguishes Pearl blankets from Friston blankets. Essential critical reading.

[9] **Ramstead, M. J. D., Sakthivadivel, D. A. R., Heins, C., et al. (2023).** "On Bayesian Mechanics: A Physics of and by Beliefs." *Interface Focus*, 13(3), 20220029.
Foundational paper for Bayesian mechanics. Establishes the mathematical framework for a "physics of beliefs."

[10] **Da Costa, L., Friston, K., Heins, C., & Pavliotis, G. (2021).** "Bayesian Mechanics for Stationary Processes." *Proceedings of the Royal Society A*, 477(2256), 20210518.
Rigorous treatment of the conditions under which internal states parameterize beliefs about external states. Partially addresses critics.

[11] **Millidge, B., Tschantz, A., & Buckley, C. L. (2021).** "Whence the Expected Free Energy?" *Neural Computation*, 33(2), 447–482.
Derives the EFE from multiple starting points. Clarifies the relationship between different formulations.

[12] **Sajid, N., Ball, P. J., Parr, T., & Friston, K. (2021).** "Active Inference: Demystified and Compared." *Neural Computation*, 33(3), 674–712.
Accessible comparison of active inference with RL and Bayesian RL. Good entry point for ML researchers.

[13] **Pezzulo, G., Rigoli, F., & Friston, K. (2018).** "Hierarchical Active Inference: A Theory of Motivated Control." *Trends in Cognitive Sciences*, 22(4), 294–306.
Develops hierarchical active inference for temporal abstraction and motivated behavior. Connects to options in hierarchical RL.

[14] **Fountas, Z., Sajid, N., Mediano, P., & Friston, K. (2020).** "Deep Active Inference Agents Using Monte-Carlo Methods." *NeurIPS 2020.*
Combines deep generative models with active inference and MCTS. Shows competitive performance on challenging tasks.

[15] **Champion, T., Bowman, H., & Grünewälder, S. (2022).** "Branching Time Active Inference with Bayesian Filtering." *Neural Computation*, 34(10), 2132–2144.
Extends deep active inference with explicit tree search. Connects sophisticated inference to practical algorithms.

[16] **Ramstead, M. J. D., Da Costa, L., et al. (2024–2025).** Recent preprints on Bayesian mechanics, path-integral formulations, and connections to gauge theory.
Cutting-edge work extending Bayesian mechanics. Check arXiv for the latest developments. The field is moving fast.

[17] **Jaynes, E. T. (2003).** *Probability Theory: The Logic of Science.* Cambridge University Press.
Not an active inference reference per se, but essential background for understanding the MaxEnt foundations of Bayesian mechanics. Jaynes' philosophy of probability underpins much of the FEP's conceptual framework.

Summary: The Big Picture

Active inference and the Free Energy Principle offer a mathematically coherent framework that unifies perception, action, learning, and planning under a single objective: minimizing (expected)

free energy. The key equations are:

$$\text{Variational free energy:} \quad \mathcal{F} = \mathrm{D}_{\mathrm{KL}}[q(\theta) \| p(\theta)] - \mathbb{E}_q[\ln p(o \mid \theta)] \qquad \text{(Complexity} - \text{Accuracy)}$$

$$\text{Expected free energy:} \quad \mathcal{G}(\pi) = -\text{Information Gain} - \text{Pragmatic Value} \qquad \text{(Exploration + Exploitation)}$$

$$\text{Policy selection:} \quad p(\pi) \propto \exp(-\gamma\, \mathcal{G}(\pi)) \qquad \text{(Softmax policy)}$$

The framework is not without its critics, and the ML community should engage with both its strengths (principled exploration, unified objective, rich mathematical structure) and its limitations (scalability, falsifiability, Markov blanket controversies). As agentic AI systems grow more sophisticated, the ideas from active inference—particularly the epistemic drive, the generative model-centric perspective, and the unification of inference and control—are likely to play an increasingly important role.

These notes were prepared for the ETH Zurich ML Lecture Series, Spring 2026.
For corrections or suggestions, please contact the course staff.